

Özgeçmiş ve Eserler Listesi

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İletişim Bilgileri

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Eğitim Bilgileri

M.Sc., Bilgisayar Bilimi, 2021, **Mila - Quebec AI Institute, Université de Montréal**, Kanada
Tezi: Identifying Electrons with Deep Learning Methods

Ph.D., Fizik, 2008, **University of Florida**, Gainesville, Florida.
Tezi: Quantum Gravitational Correction to Scalar Field Equations During Inflation

M.Sc., Fizik, 2002, **Orta Doğu Teknik Üniversitesi**, Ankara, Türkiye.
Tezi : Higher dimensional metrics of colliding gravitational plane waves

B.S., Fizik, 2000, **Orta Doğu Teknik Üniversitesi**, Ankara, Türkiye.

Araştırma Alanları

Makina Öğrenmesi, Yapay Zeka, Kozmoloji, Alternatif Gravitasyon Modelleri, Kuantum Alan Teorileri

İş Geçmişi

Misafir Araştırmacı, Mila - Quebec AI Institute, 2019-2021.

Öğretim Üyesi, İstanbul Teknik Üniversitesi, 2012-.

Doktora Sonrası Araştırmacı, Friedrich-Schiller-University of Jena, 2009-2012.

Doktora Sonrası Araştırmacı, Koç Üniversitesi, 2008-2009.

Öğretim Görevlisi, University of Florida, 2002-2008.

Öğretim Görevlisi, Orta Doğu Teknik Üniversitesi, 2000-2002.

Ödüller

Mustafa Parlar Araştırma Teşvik Ödülü 2017.

Tübitak Bilim Teşvik Ödülü 2016.

Türkiye Bilim Akademisi Üstün Başarılı Genç Bilim İnsanı Ödülü TÜBA-GEBİP-2015.

2015 Yılın Bilim İnsanı Ödülü, Bilim Kahramanları Derneği.

“Astroparticle Physics” Dergisinin En Değerli Hakemi (2010).

Araştırma Projeleri

TÜBİTAK 1001, “Carrollien Alan Teorileri”
E. O. Kahya (PI) (2026-2028).

Yapay Öğrenmede Optimizasyon Yöntemleri,
Yüksek Öğretim Kurumları Destekli Proje, E. O. Kahya (PI) (2019-2023)

IVADO Project, Machine learning for the analysis of the Large Hadron Collider Data at CERN,
Yoshua Bengio (PI), Jean-Francois Arguin (LR), E. O. Kahya (Araştırmacı) (2019-2021)

TÜBİTAK 1001, “Enflasyon ve Süpersimetrik Atraktörler,”
E. O. Kahya (PI), Proje No: 117F102, Desteklenen Bütçe: 264,400 TL, (2017-2019).

TÜBİTAK 1001, “İlk Dönem Evrendeki Kuantum Etkileri,”
E. O. Kahya (PI), Proje No: 112T817, Desteklenen Bütçe: 148,220 TL (2013-2015).

Avrupa Birliği, Marie Curie International Reintegration Grants,
E. O. Kahya (PI), Proje No: IRG-247803, Desteklenen Bütçe: \$ 137,000 (2009-2012).

Yayınlar

1. “Stationary solutions in the small c -expansion of GR”, E. Bal, E. Ekiz, E. O. Kahya, U. Zorba, arXiv:2604.23677 (2026).
2. “Do equation of state parametrizations of dark energy faithfully capture the dynamics of the late universe?”, Ö. Akarsu, M. Caruana, K. F Dialektopoulos, L. A. Escamilla, E. O. Kahya, J. L. Said, arXiv:2604.12987 (2026).
3. “Hints of sign-changing scalar field energy density and a transient acceleration phase at $z \sim 2$ from model-agnostic reconstructions”, Ö. Akarsu, M. Caruana, K. F Dialektopoulos, L. A. Escamilla, E. O. Kahya, J. L. Said, arXiv:2602.08928 (2026).
4. “The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics” E. Valentino et. al., Physics of the Dark Universe **49**, (2025) 101965.
5. “Quantization of Carrollian Fermions” E. Ekiz, E.O. Kahya, U Zorba, Phys. Rev. **D111** (2025) 105019.
6. “Holographic boundary conformal field theory within Horndeski gravity” Santos et. al., Journal of High Energy Physics **12** (2024) 1-30.
7. “On the Classification and Feature Relevance of Multiband Light Curves,” Fatma Kuzey Edes-Huyal, Zehra Cataltepe and Emre O. Kahya, AJ, **161** (2021) 004.
8. “Superconformal generalizations of auxiliary vector modified polynomial $f(R)$ theories,” S. Boran, E.O. Kahya, N. Ozdemir, M. Ozkan, U. Zorba, JCAP **04** (2020) 005.
9. “Constraints on differential Shapiro delay between neutrinos and photons from IceCube-170922A,” Sibel Boran, Shantanu Desai and E.O. Kahya, Eur. Phys. J. **C79** (2019) 3, 185.
10. “GW170817 Falsifies Dark Matter Emulators,” Sibel Boran, Shantanu Desai, Emre Kahya and Richard Woodard, Phys. Rev. **D97** Rapid Communication (2018) 041501.
11. “Loop Corrections to Primordial Non-Gaussianity,” S. Boran and E.O. Kahya, Phys. Rev. **D97** (2018) 043507.
12. “Galactic Shapiro Delay to the Crab Pulsar and limit on Einstein’s Equivalence Principle Violation,” Shantanu Desai, Emre O. Kahya, Eur. Phys. J. **C78** (2018) 2, 86.
13. “One loop corrected conformally coupled scalar mode equations during inflation,” Sibel Boran, E.O. Kahya and Sohyun Park, Phys. Rev. **D96** (2017) 105003.

14. “Quantum gravity corrections to the conformally coupled scalar self-mass-squared on de Sitter II: kinetic-conformal cross terms,” Sibel Boran, E.O. Kahya and Sohyun Park, Phys. Rev. **D96** (2017) 025001.
15. “Broken Scale Invariance, Alpha-Attractors and Vector Impurity,” O. Akarsu, S. Boran, E.O. Kahya, N. Ozdemir and M. Ozkan, Eur. Phys. J. **C77** (2017) 5, 306.
16. “Constraints on frequency-dependent violations of Shapiro delay from GW150914,” S. Desai and E.O. Kahya, Phys. Lett.**B756** (2016) 265.
17. “The Galactic One-Way Shapiro Delay to PSR B1937+21,” S. Desai and E.O. Kahya, Mod. Phys. Lett. **A31** (2016), 1650083
18. “Constructing an inflaton potential by mimicking modified Chaplygin gas,” E.O. Kahya, B. Pourhassan and S. Uraz, Phys. Rev. **D92** (2015) 103511.
19. “The universe dominated by the extended Chaplygin gas,” E.O. Kahya and B. Pourhassan, Mod. Phys. Lett. **A30** (2015), 1550070
20. “Higher order corrections of the extended Chaplygin gas cosmology with varying G and Λ ,” E.O. Kahya, M. Khurshudyan, B. Pourhassan, R. Myrzakulov, A. Pasqua, Eur. Phys. J. **C75** (2015) 2, 43.
21. “Quantum gravity corrections to the conformally coupled scalar self-mass-squared on de Sitter background,” Sibel Boran, E.O. Kahya and Sohyun Park, Phys. Rev. **D90** (2014) 124054.
22. “Observational constraints on the extended Chaplygin gas inflation,” E.O. Kahya and B. Pourhassan, Astrophysics and Space Science **353**, 677-682 (2014).
23. “Interacting two-component fluid models with varying EoS parameter,” M. Khurshudyan, B. Pourhassan and E.O. Kahya, International Journal of Geometric Methods in Modern Physics **11** 1450061 (2014).
24. “Testing a Dilaton Gravity Model using Nucleosynthesis,” S. Boran and E. O. Kahya, Advances in High Energy Physics, 282675 (2014).
25. “FRW cosmology with the extended Chaplygin gas,” B. Pourhassan and E.O. Kahya, Advances in High Energy Physics, 231452 (2014).
26. “Extended Chaplygin gas model,” B. Pourhassan and E.O. Kahya, Res. in Phys., **4** 101 (2014).
27. “The Coincidence Limit of the Graviton Propagator in de Donder Gauge on de Sitter Background,” E.O. Kahya, S. P. Miao and R.P. Woodard, J. Math. Phys. **53** (2012) 022304.
28. “A useful guide for gravitational wave observers to test modified gravity models,” E. O. Kahya, Phys.Lett.**B701** (2011) 291.
29. “The Zeta-Zeta Correlator Is Time Dependent,” E.O. Kahya, V.K. Onemli and R.P. Woodard, Phys.Lett.**B694** (2010) 101.
30. “A Completely Regular Quantum Stress Tensor with $w < -1$,” E.O. Kahya, V.K. Onemli and R.P. Woodard, Phys. Rev. **D81** (2010) 023508.
31. “A Decisive test to confirm or rule out the existence of dark matter emulators using gravitational wave observations,” E. O. Kahya, Class Quantum Grav. **25** (2008) 184008.
32. “Reduced time delay for gravitational waves with dark matter emulators,” S. Desai, E. O. Kahya and R. P. Woodard, Phys. Rev. **D77** (2008) 124041.
33. “Scalar Field Equations from Quantum Gravity during Inflation,” E. O. Kahya and R. P. Woodard, Phys. Rev. **D77** (2008) 084012.
34. “Quantum Gravity Corrections to the One Loop Scalar Self-Mass during Inflation,” E. O. Kahya and R. P. Woodard, Phys. Rev. **D76** (2007) 124005.

35. “ A Generic Test of Modified Gravity Models which Emulate Dark Matter,”
E. O. Kahya and R. P. Woodard, Phys.Lett.**B652** (2007) 213.
36. “ Quantum Instability of a $w < -1$ phase of Cosmic Acceleration,”
E. O. Kahya and V. K. Onemli, Phys. Rev. **D76** (2007) 043512.
37. “ One Loop Corrected Mode Functions for SQED during Inflation,”
E. O. Kahya and R. P. Woodard, Phys. Rev. **D74** (2006) 084012.
38. “ Charged Scalar Self-Mass during Inflation,”
E. O. Kahya and R. P. Woodard, Phys. Rev. **D72** (2005) 104001.
39. “ Higher dimensional metrics of colliding gravitational plane waves,”
M. Gurses, E. O. Kahya, A. Karasu, Phys. Rev. **D66** (2002) 024029.

Hakemlik Yaptığı Dergiler

Physical Review D, Physics Letters B, Classical and Quantum Gravity, Astroparticle Physics.

Davetli Konuşmacı Olarak Katıldığı Kurultaylar

1. Hot topics in Modern Cosmology Spontaneous Workshop XI, Korsika, Fransa, Mayıs 1-6 2017.
Sunumu: “Loop effects to Non-Gaussianity”.
2. Utrecht Cosmology Symposium 2016, Utrecht, Hollanda, 27 Haziran - 1 Temmuz 2016.
Sunumu: “Loop Corrections to Primordial non-Gaussianities”.
3. Ankara YEF Günleri 2015, 13 Şubat 2015. Sunumu: “Kozmolojik Tedirgemeler”.
4. Feza Gürsey Yaz Okulları, Eylül 1-12, 2014. Sunumu: “İlkel çekirdek sentezlenmesi”.
5. 12th Workshop on Quantization, Dualities and Integrable Systems, Koç Üniversitesi, İstanbul, Türkiye, Nisan 20-22, 2012. Sunumu: “Power Spectrum - Can it really be time-dependent?”.
6. Physics of de Sitter Spacetime , Albert Einstein Institute, Hannover, Almanya, Eylül 11-14, 2012.
Sunumu: “Quantum Effects to Scalar Particles during Inflation”.
7. IZYE 2012, İzmir, Türkiye, Haziran 19-23, 2012.
Sunumu: “Inflation Süresindeki Kuantum Etkileri”.
8. 11th Workshop on Quantization, Dualities and Integrable Systems, Pamukkale, Türkiye, Nisan 21-23, 2012. Sunumu: “Quantum effects during Inflation”.
9. Annual Meeting GRK 1523: Quantum and Gravitational Fields, Oppurg, Almanya, Mart 23-24, 2012.
Sunumu: “Realizing quantum effects during Inflation”.
10. Quantum Gravity: from UV to IR, CERN, Cenevre, İsviçre, Eylül 5-23, 2011.
Sunumu: “Effects of Loops during Inflation” .
11. Modified Gravity Approaches to the Dark Sector, Strasburg, Fransa, Haziran 28 - Temmuz 1, 2010.
Sunumu: “Testing new physics with gravitational waves”.
12. Gravitational Wave Tests of Alternative Theories of Gravity in the Advanced Detector Era, Milwaukee, Wisconsin, ABD, Haziran 26-27, 2010.
Sunumu: “Externally Triggered GW Searches and Tests of Alternate Gravity Models”.
13. Interactions in the dark: Physics of Dark Energy - Dark Matter Interactions, Leiden, Hollanda, Nisan 4-9, 2009. Sunumu: “Testing Alternative Gravity Models”.

Davetli Konuşmacı Olarak Verilen Seminerler ve Katıldığı Konferanslar

1. **GR21**, 21st International Conference on General Relativity and Gravitation, Columbia University, New York, ABD, Temmuz 10-15, 2016.
Sunumu: “Time-Dependent Scalar Mode Functions effect to Non-Gaussianity”
2. **Planck 2016**, 19th International Conference From the Planck Scale to the Electroweak Scale, Valensiya, İspanya, Mayıs 23-27 2016. Sunumu: “Zeta-Zeta propagator: Is it time-dependent?”
3. Pusan National University, Busan, Güney Kore, 27 Ağustos 2015.
Davetli Olarak Sunumu: Zeta Zeta Correlator is Time-Dependent
4. CosKASI (Cosmology Group at KASI) Korea Astronomy and Space Science Institute, Daejeon, Güney Kore, 25 Ağustos 2015.
Davetli Olarak Sunumu: Zeta Zeta Correlator is Time-Dependent
5. **Planck 2015**, 18th International Conference From the Planck Scale to the Electroweak Scale, Ioannina, Yunanistan, Mayıs 25-29 2015. Sunumu: “Zeta-Zeta propagator: Is it time-dependent?”
6. University of Granada, Granada, İspanya, Nisan 2015.
Davetli Olarak Sunumu: “Quantum effects during Inflation”
7. Orta Doğu Teknik Üniversitesi, Ankara, Türkiye, Kasım 2014.
Davetli Olarak Sunumu: “Quantum Gravitational Effects during the Early Universe”
8. The 18th annual International Conference on Particle Physics and Cosmology, **Cosmo 2014**, Kavli Institute for Cosmological Physics (KICP), Chicago, ABD, Ağustos 2014
Sunumu: “Quantum Effects on Conformal Scalars during Inflation”
9. Dark Matter in the Universe and Universal Properties of Galaxies: Theory and Observations Paris, Fransa, Haziran 8-11, 2010.
Davetli Olarak Sunumu: “Externally Triggered GW Searches and Tests of Alternate Gravity Models”
10. **12th** Gravitational Wave Data Analysis Workshop, MIT, Cambridge, MA, ABD, Aralık 13-16, 2007.
Sunumu: “Testing Modified Gravity Models by Gravitational Wave Observation”
11. **Inaugural Conference**, Penn State University, University Park, PA, ABD, Ağustos 2007.
Sunumu: “A Generic Test of Modified Gravity Models which Emulate Dark Matter”
12. University of Crete, Heraklion, Girit, Ağustos 2006.
Davetli Olarak Sunumu: “Quantum Corrected Mode Functions for SQED during Inflation”
13. **MG11**, Eleventh Marcel Grossmann Meeting of General Relativity, Berlin, Almanya, Temmuz 2006.
Sunumu: “Effective Field Equations for Charged Scalar during Inflation”

Basında

- Physics World, “Happy Dark Matter Day,” October 31 2017.
- Smithsonian Magazine, “What the Neutron Star Collision Means for Dark Matter,” October 31 2017.
- MIT Technology Review, “The Best of the Physics arXiv,” October 28 2017.
- Forbes, “Merging Neutron Stars Deliver Deathblow To Dark Matter And Dark Energy Alternatives,” October 25 2017.
- Ars Technica, “Colliding neutron stars apply kiss of death to theories of gravity,” October 25 2017.
- Alphr, “The neutron star merger is good news for dark matter,” October 25 2017.
- New Scientist, Particle race could settle dark matter debate, 7 Mart 2008.